UAV Trajectory Control Against Hostile Jamming in Satellite-UAV Coordination Networks

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***Abstract*—Large unmanned aerial vehicle (UAV) performs reconnaissance and data collection missions against hostile jamming in satellite-UAV coordination Networks. In this case, the ground base station (BS) is unable to provide access service to the UAV, thus the UAV has to rely on information support from the satellite communication system. Low earth orbit (LEO) satellites provide access beams for UAVs, and the UAV transmits the collected data to the satellite via uplink. Due to the unknown and uncertain environment, it is difficult for large UAV to get an effective planned flight trajectory, and the presence of malicious jamming further exacerbates the complexity of trajectory control. To address this problem, a reinforcement learning (RL) based trajectory control approach is proposed to explore the unknown jamming environment and realize autonomous trajectory planning. Finally, the simulation results prove the effectiveness of the proposed approach.**

Keyworks—Anti-jamming, UAV trajectory control, Reinforcement learning.

I. Introduction

Due to the advantages of fast deployment and low capital cost, UAV is believed to be a potentially promising choice to data collection and emergency communication [1]. But the UAV needs the assistance of the BS during the mission execution. BS could provide access channel to give trajectory control information, and UAVs also need to offload the collected data back to the BS in time, to save its own cache space for the next task. In the hostile jamming environment, the ground BS is unable to provide access service to the UAV, thus the UAV has to rely on information support from the satellite communication system. LEO satellite communication system is an irreplaceable safeguard in the confrontation environment [2], [3]. LEO satellites provide access beams for UAVs, and the UAV transmits the collected data to the satellite via uplink. Due to their own limitations, UAVs and satellites are able to coordinate with each other, leading to a cognitive satellite-UAV network [4]. The coordination between satellites and

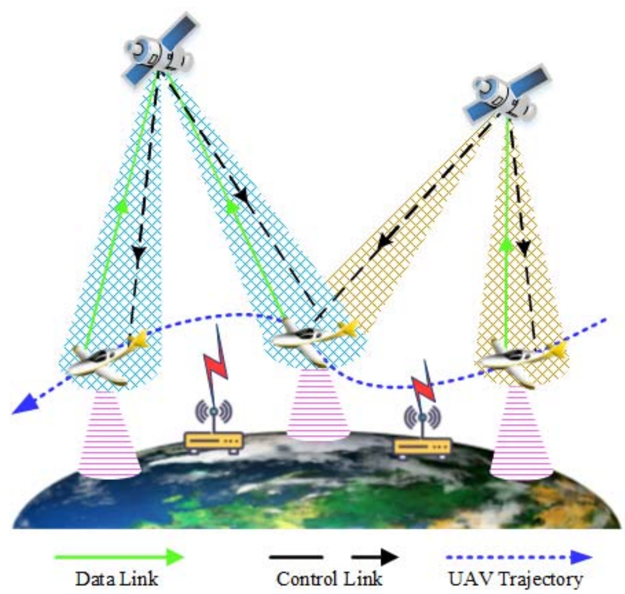

Fig. 1: System model.

UAVs has drawn increasing attentions. The authors in [5] studied a satellite-UAV mobile edge caching IoT system, and UAVs were used to enhance the data cashing ability. UAV's trajectory design is an important issue, which is closely related to the UAV's kinematic parameters [6]. With the aid of BS or satellite, the UAV can realize efficient multi-point data acquisition through trajectory planning. The authors in [7] studied the 3D deployment for multiple UAVs acting as flying BS. The authors in [8] investigated the multi-UAV communication scheduling and joint optimization of trajectory and power control. The authors in [9] proposed a decoupled heuristic solution to minimize the total hovering and traveling time of UAV. The UAV trajectory optimization were studied in

[10] which could meanwhile exploit the map and quickly learn the propagation parameters. In [11], the authors investigated UAV trajectory control problem at the edges of adjacent cells to offload data for BSs.

However, these works mainly solved the trajectory optimization by convex optimization techniques and centralization optimization rather than distributed and automatical adjustment. But global information is sometimes difficult to obtain, especially in the uncertain jamming environment. The joint optimization problem considering the anti-jamming requirement are rare investigated in the state-of-the-art as well. On the other hand, beam switching is also a condition that seriously affects UAV trajectory planning. Because UAVs cannot access the area with a poor beam quality. Thus, the beam selection and trajectory control should be optimized jointly rather than in a separated manner, especially in the anti-jamming satellite-UAV coordination networks [8].

Different from the existing works [12], which makes pre-deployment before flight, in this paper, the trajectory control is accomplished automatically. In this paper, the automatical trajectory control problem for UAV in the anti-jamming satellite-UAV coordination networks is formulated as a Markov decision process problem (MDP). To address this problem, an RL based trajectory control approach is proposed to explore the unknown jamming environment and realize autonomous trajectory planning. The remaining part of this paper is organized as follows. System model and problem formulation are shown in Section II. The proposed anti-jamming trajectory control algorithm is elucidated in Section III. Section IV shows simulation results and Section V draws the conclusions.

II. System Model and Problem Formulation

As indicated in Fig. 1, UAV needs to go from one data collection point to another, and the data collected from the previous point needs to be offloaded to the satellite via the uplink during flight, to save storage space for the next data collection.

A. Satellite coverage model

The orbit height of LEO satellite is H. The earth radius is $R_E = 6371\text{km}$, and the minimum elevation angle from ground observation point to satellite is $\theta = 10°$. By simple geometric calculation, the geocentric angle between a satellite and a ground observation point is $\alpha = \arccos\left[\cos\theta \times R_E/\left(R_E+H\right)\right] - \theta$. The angular velocity of satellite are $w = \sqrt{GM/\left(R_E+H\right)^3}$, where G and M are the Gravitational constant and Earth's mass. The longest continuous coverage time of a satellite to an UAV is $T = 2\alpha/w$. The square beam is used to replace the circular beam for simplified calculation. Each satellite has $N_B \times N_B$ beam squares. Assuming that the tracking service of the beam to the UAVs is not considered, the beam cell sweeps the coverage area at high speed, and the UAV needs to switch quickly and frequently between the different beam cell or even different satellites. The switching time per beam is $T_b = T/N_B$.

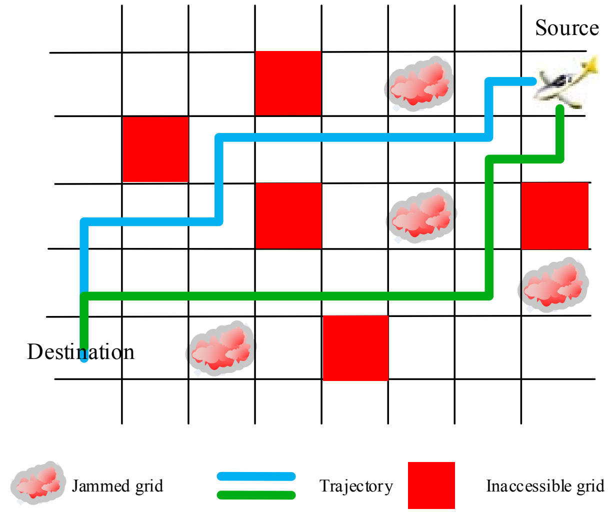

Fig. 2: Problem formulation.

B. UAV motion model

The UAV has the capacity of environmental awareness, and can detect the surrounding environment. The collected reconnaissance data and current geographic location are transmitted to the satellite via the uplink, and the satellite transmits the nearby beam information and trajectory control information to the UAV via the downlink.

Without loss of generality, we consider a 3D coordinate system where the large UAV flies at a fixed altitude H_u above ground. As shown in Fig. 2, the Manhattan network model is used to model the target region. The coordinates projected on the horizontal plane of the current location and next location are defined as $l_t = [x_t, y_t]$ and $l_{t+1} = [x_{t+1}, y_{t+1}]$. UAV's speed is v_u, and UAV can only choose four motion selection: forward, backward, left, right, then, goes into the adjacent grid areas, i.e., $\|l_t - l_{t+1}\| \leq 1$, where $\|l_t - l_{t+1}\|$ represents the Manhattan distance from l_t to l_{t+1}.

$$\|l_t - l_{t+1}\| = |x_t - x_{t+1}| + |y_t - y_{t+1}|\,. \tag{1}$$

C. Channel transmission model

The LoS link between the satellite and UAV is mainly considered in this paper. What's more, the Doppler effect due to the satellite movement is assumed to be perfectly compensated. According to [13], the shadowed-Rician model is used to formulate the satellite link, the probability density function of $X_1{=}|h_i|^2$ is expressed as:

$$f_{X_1}(x) = \alpha_{n,i} \exp\left(-\beta_{n,i}x\right) {}_1F_1\left(m_{n,i}; 1; \delta_{n,i}x\right), \tag{2}$$

and Nakagami-m fading mdel is utilized to formulate the terrestrial jamming links, and the probability density function of $X_2{=}|h_J|^2$ is defined as:

$$f_{X_2}(x) = \frac{x^{\vartheta_{n,i}-1}\exp(-\frac{x}{\eta_{n,i}})}{\Gamma(\vartheta_{n,i})\eta_{n,i}{}^{\vartheta_{n,i}}}. \tag{3}$$

Therefore, the maximum transmission rate of the satellite uplink is defined as:

$$C_i = B_i \log_2(1+r_i), \quad r_i = \frac{h_i d^{-\beta_u} P_{u,i}}{h_J d_J^{-\beta_J} P_J + \sigma^2}, \tag{4}$$

where r_i denotes the signal-plus-jamming-to-noise ratio (SJNR) at the satellite receiver from the i-th grid, and $P_{u,i}$, P_J, σ^2, h, β respectively represent the UAV transmitting power, jamming power, channel noise power, small-scale and large-scale channel fading coefficient, respectively. B_i represents the access channel bandwidth provided by a satellite beam covering the current region. When there is no channel remaining in the beam cell, i.e., $B_i = 0$, or the channel fading of this beam is very severe, i.e., $r_i < r_{th}$, the UAV cannot enter this grid, i.e., $\delta(r_i \geq r_{th}) = 0$. Otherwise, this beam could provide available service, and $\delta(r_i \geq r_{th}) = 1$.

D. Problem formulation

As shown in Fig. 2, the motion direction of the UAV is parallel or perpendicular to the direction of satellite movement. Each grid represents a beam cell, which will be covered by another beam at every T_b. The total energy consumption of the UAV includes two components. The first one is the propulsion energy, which is required for supporting its mobility. The other component is the data offloading energy, which is mainly related to the offloading data traffic and corresponding channel state. The propulsion energy consumption of the UAV is:

$$S_v = e_1 L, \tag{5}$$

where e_1 is the unit flying energy consumption, and L is the Manhattan distance that the UAV passes through. In order to reach the next collection point in time, half of the energy reserve is used to ensure the flight distance, thus, UAV can fly up to L_{th}. The data offloading energy consumption is:

$$S_{c,i} = e_2 D_i, \quad D_i = C_i T_i, \tag{6}$$

where D_i is the offloading data traffic at i-th grid, and e_2 is the unit collecting energy consumption per bit.

Due to the declining beam quality or the jamming impact, UAV needs to frequently switch beam grid and get into the grid with better channel state. The problem studied in this paper is that the UAV needs to offload as much data as possible through the trajectory control and beam switching with the limitation of energy reserve E_o in a point to point flight:

$$P1: \max_{\xi=[l_s, l_2 \cdots l_i \cdots, l_d]} \sum_i D_i \delta(r_i \geq r_{th}), \quad s.t. \ \sum_i S_{c,i} + S_v \leq E_o \tag{7}$$

where $\xi = [l_s, l_2 \cdots l_i \cdots, l_d]$ is the UAV trajectory from source l_s to the destination l_d.

Algorithm 1:: Learning-based Anti-jamming Trajectory Control Algorithm (LATC)

Input: The source location l_s, and the destination location l_d.

1: Initialize $Q_t(s_t, a_n)$=0, $\pi_n(t)$=1/4.
2: **for** *epoch* $= 1$ **to** K **do**
3: $t = 1$, $\xi = [l_s]$.
4: **while** $s_t \neq l_d$ & $t \leq L_{th}$ **do**
5: In the current location s_t, UAV obtains the available subset of actions $A' \subset A$.
6: UAV chooses action a_t from A' according to $\pi_n(t)$, and gets into the next location s_t', and updates $\xi = [l_s, \ldots, s_t, s_t']$.
7: UAV offloads data traffic in the next location s_t' and obtains the reward r_t.
8: UAV updates $Q_{t+1}(s_t, a_n)$ and $\pi_n(t)$.
9: $t \leftarrow t+1$, $s_t \leftarrow s_t'$.
10: **end while**
11: If a feasible trajectory is found, each selected action on the trajectory will be given a delayed return R_0 to their Q value.
12: **end for**

Output: UAV trajectory ξ, and the total unloading data $\sum_i D_i$.

III. Learning-based Anti-jamming Trajectory Control Algorithm

Due to the unknown and uncertain jamming environment, the trajectory control for large UAV is difficult to address by the traditional approaches, such as convex optimization theory which is widely used in the existing works. Therefore, an RL paradigm is proposed in this paper to deal with the exploration with the unknown jamming environment and search for the optimal possible trajectory.

In this paper, the target area is meshed and the path planning is carried out in a discrete way. As mentioned earlier, it is also a requirement for satellite beam switching. In this case, the selection of the next grid is not only required by trajectory planning, but also limited by beam switching. Meanwhile, it is only relevant to the current grid and the available action space, not to the previous selection. Thus, it is an MDP problem. The anti-jamming trajectory control problem could be modeled as a 4-tuple MDP, with state space S, action space A, reward function R, and dynamic transition probability Pr [14]. However, it is hard to get the accurate Pr due to the unknowability of the hostile jamming environment. Thus, a reinforcement learning based anti-jamming trajectory control algorithm is proposed to look for the best possible trajectory. The state space is defined as the current location:

$$\{s_t \in S \,|S = l_1, l_2, \ldots, L_N\}, \tag{8}$$

where $N = X \times Y$ is the number of all the grids, and X and Y are the corresponding two-dimensional side length. The action space in this paper is defined as:

$$\{a_t \in A \,|A = a_1, a_2, a_3, a_4\}, \tag{9}$$

where a_1, a_2, a_3, a_4 are expressed as the action of “up”, “down”, “left”, “right”. The reward function is defined as the

TABLE I: Shadowed-Rician model parameters.

Shadowed-Rician model options	b	m	ω
Heavy shadowing model	0.063	0.739	8.97×10^{-4}
Average shadowing model	0.126	10.1	0.835
Light shadowing model	0.316	38.8	2.58

TABLE II: Jamming link model parameters.

Jamming link status	ϑ	η
Heavy jammed Nakagami-m model	4	0.5
Average jammed Nakagami-m model	2	0.5
Light jammed Nakagami-m model	1	0.5

traffic data offloaded in the current grid:

$$r_t = T_t C_t, \tag{10}$$

where T_t and C_t are the offloading time and offloading rate.

The Q function of the UAV is expressed as $Q(s_t, a_t)$, which is updated as follows:

$$\begin{aligned} Q_{t+1}(s_t, a_t) = (1-\alpha)\, Q_t(s_t, a_t) \\ + \alpha \left(r_t + \eta \max_{a_t{}'} Q_t(s_t{}', a_t{}') \right) \end{aligned}, \tag{11}$$

where $s_t{}'$ is the next state after performing action a_t, and $a_t{}'$ is next possible action to perform. $\pi(t)$= $[\pi_1(t), \pi_2(t) \ldots \pi_4(t)]$ is the mixed strategies of action selection, and $\pi_n(t)$ represents the probability to choose the action $a_n \in [a_1, a_2, a_3, a_4]$. Specially, the $\pi_n(t)$ is given by:

$$\pi_n(t) = \frac{e^{Q_t(s_t, a_n)/\tau}}{\sum_{n'=1}^{4} e^{Q_t(s_t, a_{n'})/\tau}}, \tag{12}$$

where, τ is the parameter of the Boltzmann model.

$$\begin{array}{ll} \tau = \tau_0 e^{(-\upsilon t)} & \tau \geq \hat{\tau} \\ \tau = \hat{\tau} & \tau < \hat{\tau} \end{array}, \tag{13}$$

τ_0 is related to the exploration time, and $\hat{\tau}$ represents the ending condition in the exploration stage. υ affects the transition from exploration to exploitation. The learning-based anti-jamming trajectory control algorithm is summarized in Algorithm 1.

IV. Simulation Results

Simulation experiments are carried out to prove the effectiveness of the proposed approach. Each grid has three shading model options which are shown in Table I, and there are three jamming link status which are shown in Table II. The other parameters are $H = 800$km, $T_b = 0.4483$min, $\sigma = 0.1$, $\beta = 2$, $B = 1$MHz, $p_u = p_J = 1$W, $\tau_0 = 10^7$, $\hat{\tau} = 0.1$, $\eta = 0.6$, $r_{th} = 1$, $L_{th} = 15$.

The trajectory from the same source and different destination are shown in Fig. 3, and the colors represent the different channel conditions. We can see that after the learning process, the environment dynamic is basically obtained by UAV, and it automatically makes the probabilistically optimal trajectory control to reach the destination via a trajectory with a better channel state. However, as shown in Fig. 4, the jamming degree and beam quality have significant influence on the trajectory control, the higher jamming degree may result in the absence of better options in the environment and greatly affects UAV's flight and data offloading. On the contrary, the higher beam quality can effectively counter the effects of jamming and provide better options for data offloading. Different RL

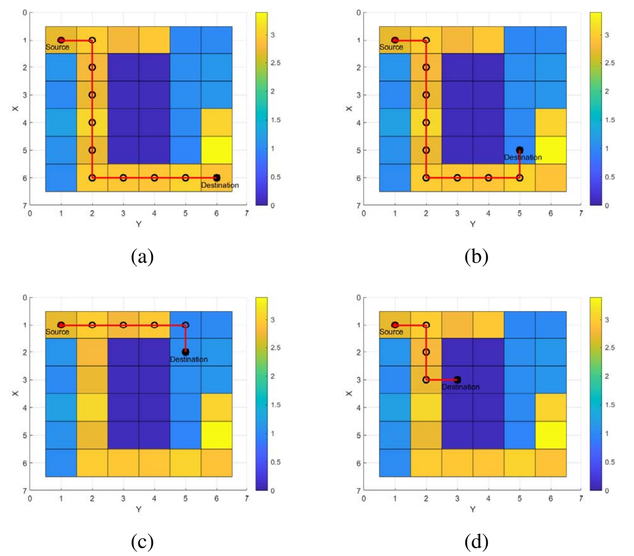

Fig. 3: The trajectory presentation and comparison.

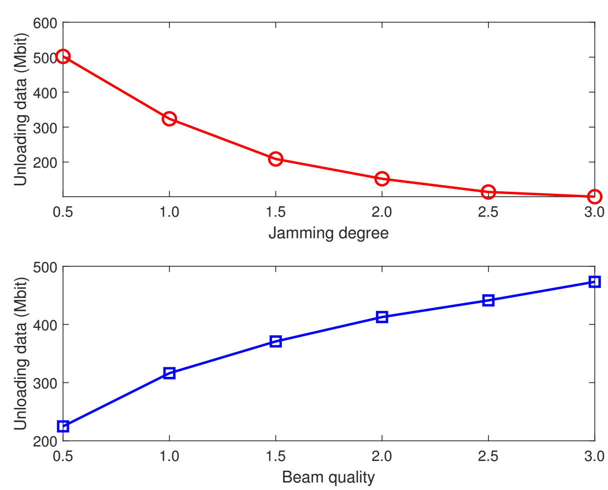

Fig. 4: The influence of jamming degree and beam quality.

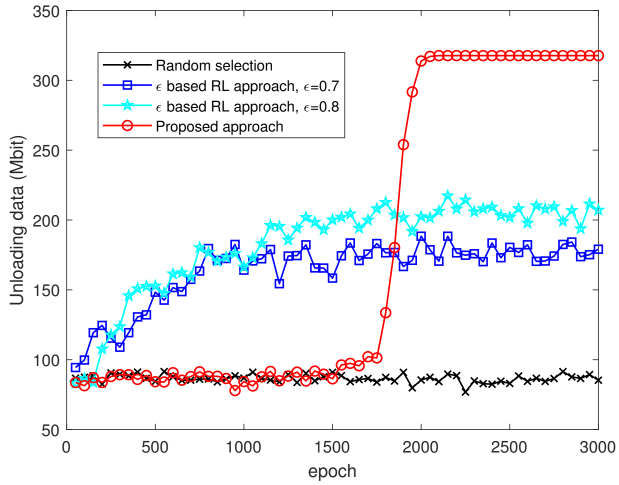

Fig. 5: The comparison of different solving approaches.

approaches are compared in this paper. As indicated in Fig. 5, the proposed algorithm has better performance in utility and convergence.

V. Conclusion

This paper investigates the anti-jamming trajectory control problem in the hostile jamming environments. LEO satellites provide access beams for UAV, and the UAV offloads current collected data to the satellite. Due to the unknown and uncertain environment, the trajectory control for UAV is difficult to get an effective planned flight trajectory, and the presence of malicious jamming further exacerbates the complexity of trajectory control. Thus, an RL based anti-jamming trajectory control approach is proposed to explore the unknown jamming environment and realize autonomous trajectory control. Finally, the simulation results prove the effectiveness of the proposed algorithm. However, there are some problem worth further studied, such as the larger grid space and high intelligence jamming, which will be investigated in the future works.

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References

[1] S. Zhang and J. Liu, "Analysis and optimization of multiple unmanned aerial vehicle-assisted communications in post-disaster areas," *IEEE Transactions on Vehicular Technology*, vol. 67, no. 12, pp. 12 049–12 060, 2018.

[2] C. Han, A. Liu, H. Wang, L. Huo, and X. Liang, "Dynamic anti-jamming coalition for satellite-enabled army iot: A distributed game approach," *IEEE Internet of Things Journal*, vol. 7, no. 11, pp. 10 932–10 944, 2020.

[3] C. Han, A. Liu, L. Huo, H. Wang, and X. Liang, "A prediction-based resource matching scheme for rentable leo satellite communication network," *IEEE Communications Letters*, vol. 24, no. 2, pp. 414–417, 2020.

[4] C. Liu, W. Feng, Y. Chen, C. Wang, and N. Ge, "Cell-free satellite-uav networks for 6g wide-area internet of things," *IEEE Journal on Selected Areas in Communications*, pp. 1–1, 2020.

[5] S. Gu, Y. Wang, N. Wang, and W. Wu, "Intelligent optimization of availability and communication cost in satellite-uav mobile edge caching system with fault-tolerant codes," *IEEE Transactions on Cognitive Communications and Networking*, pp. 1–1, 2020.

[6] J. Zhang, Y. Zeng, and R. Zhang, "Uav-enabled radio access network: Multi-mode communication and trajectory design," *IEEE Transactions on Signal Processing*, vol. 66, no. 20, pp. 5269–5284, 2018.

[7] M. Mozaffari, W. Saad, M. Bennis, and M. Debbah, "Efficient deployment of multiple unmanned aerial vehicles for optimal wireless coverage," *IEEE Communications Letters*, vol. 20, no. 8, pp. 1647–1650, Aug 2016.

[8] Q. Wu, Y. Zeng, and R. Zhang, "Joint trajectory and communication design for multi-uav enabled wireless networks," *IEEE Transactions on Wireless Communications*, vol. 17, no. 3, pp. 2109–2121, 2018.

[9] O. M. Bushnaq, A. Celik, H. Elsawy, M. Alouini, and T. Y. Al-Naffouri, "Aeronautical data aggregation and field estimation in iot networks: Hovering and traveling time dilemma of uavs," *IEEE Transactions on Wireless Communications*, vol. 18, no. 10, pp. 4620–4635, 2019.

[10] O. Esrafilian, R. Gangula, and D. Gesbert, "Learning to communicate in uav-aided wireless networks: Map-based approaches," *IEEE Internet of Things Journal*, vol. 6, no. 2, pp. 1791–1802, 2019.

[11] F. Cheng, S. Zhang, Z. Li, Y. Chen, N. Zhao, F. R. Yu, and V. C. M. Leung, "Uav trajectory optimization for data offloading at the edge of multiple cells," *IEEE Transactions on Vehicular Technology*, vol. 67, no. 7, pp. 6732–6736, 2018.

[12] X. Li, W. Feng, Y. Chen, C. Wang, and N. Ge, "Maritime coverage enhancement using uavs coordinated with hybrid satellite-terrestrial networks," *IEEE Transactions on Communications*, vol. 68, no. 4, pp. 2355–2369, 2020.

[13] K. An, M. Lin, W. Zhu, Y. Huang, and G. Zheng, "Outage performance of cognitive hybrid satellite–terrestrial networks with interference constraint," *IEEE Transactions on Vehicular Technology*, vol. 65, no. 11, pp. 9397–9404, 2016.

[14] C. Han, L. Huo, X. Tong, H. Wang, and X. Liu, "Spatial anti-jamming scheme for internet of satellites based on the deep reinforcement learning and stackelberg game," *IEEE Transactions on Vehicular Technology*, vol. 69, no. 5, pp. 5331–5342, 2020.